3-10-26 (WEEK 3)

Shun/翔海 (@shun4midx)

PROPOSITION

The probability that a uniformly random permutation is a derangement is $\sum_{r=0}^{n} \frac{(-1)^r}{r!} \xrightarrow{n\to\infty} \frac{1}{e}$

BONFERRONI INEQUALITIES

Recall: $\forall$ events E, F in a probability space, $\mathbb{P}(E\cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E\cap F) \le \mathbb{P}(E) + \mathbb{P}(F)$

PROPOSITION (UNION BOUND)

Let $E_1, E_2, \dots, E_n$ be events in a probability space. Then, $\mathbb{P}(\bigcup_{i=1}^{n} E_i) \le \sum_{i=1}^{n} \mathbb{P}(E_i)$

Proof

Let $F_i := E_i \setminus \bigcup_{j<i} E_j$

Then, $\bigcup_{j=1}^{i} F_j = \bigcup_{j=1}^{i} E_j$ (Since $F_j \subseteq E_j$ $\forall j$, thus $\bigcup_{j=1}^{i} F_j \subseteq \bigcup_{j=1}^{i} E_j$. If $x \in \bigcup_{j=1}^{i} E_j$, let j_0 be the smallest idx, s.t. $x \in E_{j_0}$. Then, $x \in F_{j_0} \subseteq \bigcup_{j=1}^{i} F_j$. $\therefore \bigcup_{j=1}^{i} E_j \subseteq \bigcup_{j=1}^{i} F_j$)

Notice, the F_j's are mutually exclusive ($\because F_i = E_i \setminus \bigcup_{j<i} E_j = E_i \setminus \bigcup_{j<i} F_j$ $\therefore F_i \cap F_j = \emptyset$ $\forall j<i$)

$\therefore \mathbb{P}(\bigcup_{i=1}^{n} E_i) = \mathbb{P}(\bigcup_{i=1}^{n} F_i) \underset{\text{additivity}}{=} \sum_{i=1}^{n} \mathbb{P}(F_i) \underset{\text{monotinicity}}{\le} \sum_{i=1}^{n} \mathbb{P}(E_i)$ $\square$

EXAMPLE (DERANGEMENTS)

Let $\pi \in S_n$ be a uniformly random permutation. Let E_i be the event that $\{\pi(i) = i\}$

Then, $\mathbb{P}(\bigcup_{i=1}^{n} E_i) \le \sum_{i=1}^{n} \mathbb{P}(E_i) = \sum_{i=1}^{n} \frac{1}{n} = 1$ (see, it's correct, but not necessarily useful all the time)

THEOREM (BONFERRONI)

Let $E_1, \dots, E_n$ be events in a probability space $(S, \mathbb{P})$. Let $1 \le t \le n$. Then,

$$\mathbb{P}(\bigcup_{i=1}^{n} E_i) \begin{matrix} \le (t:\text{odd}) \\ \ge (t:\text{even}) \end{matrix} \sum_{r=1}^{t} (-1)^{r+1} \sum_{1\le i_1<\dots<i_r\le n} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r})$$

EXAMPLE (DERANGEMENTS, t=2)

Let E_i be as before.

Then, $\underbrace{\mathbb{P}(\bigcup_{i=1}^{n} E_i)}_{\mathbb{P}(\text{NOT derangement}) = 1-\frac{1}{e}} \ge \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{1\le i<j\le n} \mathbb{P}(E_i \cap E_j) = \sum_{i=1}^{n} \frac{1}{n} - \sum_{1\le i<j\le n} \frac{(n-2)!}{n!} = 1 - \binom{n}{2}\frac{(n-2)!}{n!} = 1 - \frac{1}{2} = \frac{1}{2}$

($\pi(i)=i, \pi(j)=j$. Then permute the remaining $n-2$ elements)

PROOF OF BONFERRONI'S INEQUALITIES

By induction on n,

Base case: $t=1 \to$ union bound ✓

Induction Step: ($t \Rightarrow t+1$) As before, $\bigcup_{i=1}^{n} E_i = \bigcup_{i=1}^{n} F_i$, where $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$

Then, $\mathbb{P}(\bigcup_{i=1}^{n} E_i) = \mathbb{P}(\bigcup_{i=1}^{n} F_i) = \sum_{i=1}^{n} \mathbb{P}(F_i) = \sum_{i=1}^{n} \mathbb{P}(E_i \setminus \bigcup_{j=1}^{i-1} E_j) \overset{\text{additivity}}{=} \sum_{i=1}^{n} [\mathbb{P}(E_i) - \mathbb{P}(E_i \cap (\bigcup_{j<i} E_j))]$

$= \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{i=1}^{n} \mathbb{P}(E_i \cap (\bigcup_{j<i} E_j)) = \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{i=1}^{n} \mathbb{P}(\bigcup_{j<i} (E_i \cap E_j))$

We use the induction hypothesis for t to bound $\mathbb{P}(\bigcup_{j<i} (E_i \cap E_j))$

If t: odd, then $\mathbb{P}(\bigcup_{j<i} (E_i \cap E_j)) \le \sum_{r=1}^{t} (-1)^{r+1} \sum_{j_1<\dots<j_r} \mathbb{P}(\bigcap_{\ell=1}^{r} (E_i \cap E_{j_\ell}))$

$= \sum_{r=1}^{t} (-1)^{r+1} \sum_{j_1<\dots<j_r<i} \mathbb{P}(E_{j_1} \cap \dots \cap E_{j_r} \cap E_i)$

$\Rightarrow \sum_{i=1}^{n} \mathbb{P}(\bigcup_{j<i} (E_i \cap E_j)) \le \sum_{i=1}^{n} \sum_{r=1}^{t} (-1)^{r+1} \sum_{1\le j_1<\dots<j_r<i} \mathbb{P}(E_{j_1} \cap E_{j_2} \cap \dots \cap E_{j_r} \cap E_i)$

$= \sum_{r=2}^{t+1} (-1)^{r} \sum_{1\le j_1<\dots<j_r\le n} \mathbb{P}(E_{j_1} \cap \dots \cap E_{j_r})$

$\therefore \mathbb{P}(\bigcup_{i=1}^{n} E_i) \ge \sum_{i=1}^{n} \mathbb{P}(E_i) + \sum_{r=2}^{t+1} (-1)^{r+1} \sum_{1\le j_1<\dots<j_r\le n} \mathbb{P}(E_{j_1} \cap \dots \cap E_{j_r}) = \sum_{r=1}^{t+1} (-1)^{r+1} \sum_{1\le j_1<\dots<j_r\le n} \mathbb{P}(E_{j_1} \cap \dots \cap E_{j_r})$

Similarly, if t: even, all that changes is the direction of the ineq sign. $\square$

CONTINUITY OF PROBABILITY

Shun/翔海 (@shun4midx)

DEFINITION

Given a probability space $(S, \mathbb{P})$, we say a sequence of events $E_1, E_2, \ldots$ is increasing if $E_1 \subseteq E_2 \subseteq \ldots$, or it is decreasing if $E_1 \supseteq E_2 \supseteq \ldots$

DEFINITION

Let $E_1 \subseteq E_2 \subseteq E_3 \subseteq \ldots$ be an increasing sequence of events in a probability space. Then, we define the limit $\lim_{n\to\infty} E_n = \bigcup_{n=1}^{\infty} E_n$

Let $E_1 \supseteq E_2 \supseteq E_3 \supseteq \ldots$ be a decreasing sequence of events, then we define $\lim_{n\to\infty} E_n = \bigcap_{n=1}^{\infty} E_n$

OBSERVE

Notice, if $E_1, E_2, E_3, \ldots$ is increasing, then $E_1 \subseteq E_2 \subseteq E_3 \subseteq \ldots \subseteq E_n \subseteq \ldots \subseteq \lim_{n\to\infty} E_n$

If $E_1, E_2, E_3, \ldots$ is decreasing, then $E_1 \supseteq E_2 \supseteq E_3 \supseteq \ldots \supseteq E_n \supseteq \ldots \supseteq \lim_{n\to\infty} E_n$

PROPOSITION

If $E_1, E_2, E_3, \ldots$ is increasing or decreasing as a sequence of events in a probability space $(S, \mathbb{P})$, then $\mathbb{P}(\lim_{n\to\infty} E_n) = \lim_{n\to\infty} \mathbb{P}(E_n)$ (i.e. "probability is cont.")

Proof

For the increasing case,

For each n, let $F_n = E_n \setminus \bigcup_{i=1}^{n-1} E_i$. Then, $\forall n$, $\bigcup_{i=1}^{n} F_i = \bigcup_{i=1}^{n} E_i = E_n$.

Moreover, $\bigcup_{n=1}^{\infty} F_n = \bigcup_{n=1}^{\infty} E_n = \lim_{n\to\infty} E_n$

↳ Since $F_n \subseteq E_n$, thus $\bigcup_{n=1}^{\infty} F_n \subseteq \bigcup_{n=1}^{\infty} E_n$, and if $x \in \bigcup_{n=1}^{\infty} E_n$, let $n_0 = \min$ index s.t. $x \in E_{n_0}$, then $x \in F_{n_0} \subseteq \bigcup_{n=1}^{\infty} F_n$

By def of limit, $\mathbb{P}(\lim_{n\to\infty} E_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} E_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} F_n) \overset{\text{by countable additivity}}{=} \sum_{n=1}^{\infty} \mathbb{P}(F_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(F_i) = \lim_{n\to\infty} \mathbb{P}(\bigcup_{i=1}^{n} F_i) = \lim_{n\to\infty} \mathbb{P}(\bigcup_{i=1}^{n} E_i) = \lim_{n\to\infty} \mathbb{P}(E_n)$ □

For the decreasing case,

Notice, if $E_1 \supseteq E_2 \supseteq E_3$ is decreasing, then $E_1^c \subseteq E_2^c \subseteq E_3^c \subseteq \ldots$ is increasing

from before, we get $\lim_{n\to\infty} \mathbb{P}(E_n^c) = \mathbb{P}(\lim_{n\to\infty} E_n^c) = \mathbb{P}(\bigcup_{n=1}^{\infty} E_n^c) \overset{\text{DeM}}{=} \mathbb{P}((\bigcap_{n=1}^{\infty} E_n)^c) = 1 - \mathbb{P}(\bigcap_{n=1}^{\infty} E_n) = 1 - \mathbb{P}(\lim_{n\to\infty} E_n)$

$\therefore 1 - \lim_{n\to\infty} \mathbb{P}(E_n) = 1 - \mathbb{P}(\lim_{n\to\infty} E_n) \Rightarrow \lim_{n\to\infty} \mathbb{P}(E_n) = \mathbb{P}(\lim_{n\to\infty} E_n)$ □

DEFINITION

Given any infinite seq of events $E_1, E_2, E_3, \ldots$ in a probability space $(S, \mathbb{P})$, we define $\limsup_{n\to\infty} E_n := \lim_{n\to\infty} \underbrace{\bigcup_{i=n}^{\infty} E_i}_{\text{decreasing seq} \Rightarrow \text{well-def}} = \bigcap_{n=1}^{\infty} (\bigcup_{i=n}^{\infty} E_i)$

Similarly, we can also define $\liminf_{n\to\infty} E_n := \lim_{n\to\infty} \bigcap_{i=n}^{\infty} E_i = \bigcup_{n=1}^{\infty} (\bigcap_{i=n}^{\infty} E_i)$

INTERPRETATION

$\limsup_{n\to\infty} E_n$ = "infinitely many of the events E_n occur" (but maybe still infinitely many events do NOT occur)

$\liminf_{n\to\infty} E_n$ = "only finitely many of the events do NOT occur"

Notice, $\liminf_{n\to\infty} E_n \subseteq \limsup_{n\to\infty} E_n$ (but not vice versa)

THEOREM (THE FIRST BOREL-CANTELLI LEMMA)

Let $(S, \mathbb{P})$ be a probability space, and let $E_1, E_2, E_3, \ldots$ be a sequence of events

If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$, then $\mathbb{P}(\limsup_{n\to\infty} E_n) = 0$

Proof

By def, $\mathbb{P}(\limsup_{n\to\infty} E_n) = \mathbb{P}(\lim_{n\to\infty} \bigcup_{i=n}^{\infty} E_i)$

$= \lim_{n\to\infty} \mathbb{P}(\bigcup_{i=n}^{\infty} E_i)$ (Continuity of Probability)

$\leq \lim_{n\to\infty} \sum_{i=n}^{\infty} \mathbb{P}(E_i)$ (Union bound)

$= 0$ ($\because \sum_{n=1}^{\infty} \mathbb{P}(E_n)$ conv) □

REMARK

If $\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty$, then $\mathbb{P}(\text{infinitely many } E_n \text{ occur}) = 0 \Rightarrow \mathbb{P}(\text{only finitely many } E_n \text{ occur}) = 1$